



Information Technology Institute (ITI)

Analog IC Design Track

Summer Camp 2026

Lab 01

LPF Simulation and MOSFET Characteristics



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PART 1: Low Pass Filter Simulation (LPF)

1. Transient Analysis

- 1) Design a first order low pass filter that has $R = 1k\Omega$ and 1ns time constant.

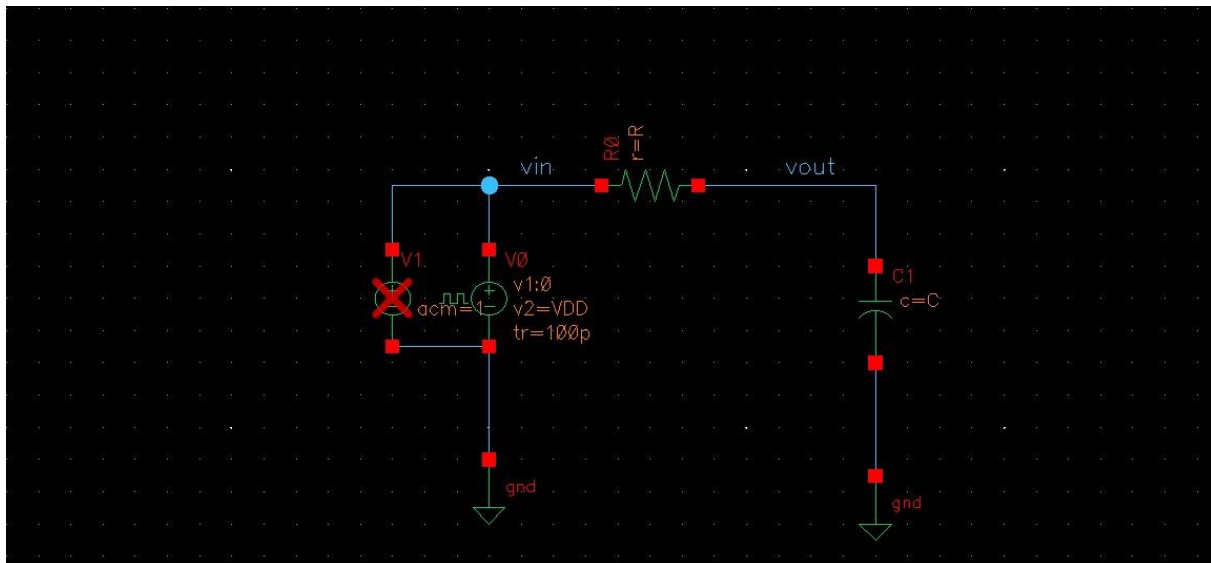


Figure 1: LPF schematic

- Using $c=0.5pF \rightarrow \tau = 0.5n s$
- Period = $T_{clk} = 52n s$ (Random)

- 2) Report transient analysis results for two periods (use max time step = $T_{clk}/100$).

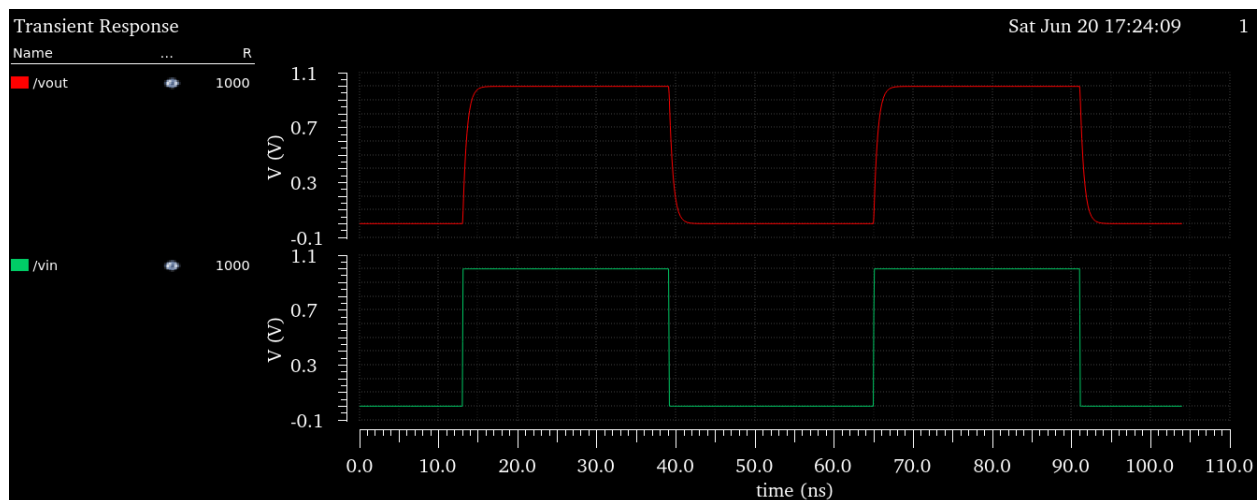


Figure 2: two periods (use max time step = $0.52ns$)



3) Calculate rise and fall time (10% to 90%) using Cadence calculator expressions. Export the expressions to adexl.

Test	Output	Nominal	Spec	Weight	Pass/Fail
lpf_part1	/vout				
lpf_part1	/vin				
lpf_part1	/V0/PLUS				
lpf_part1	/C1/PLUS				
lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	1.103n			
lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	1.101n			

Figure 3: t_{rise} & t_{fall}

3) Compare simulation with analytical results in a table

analytical	simulation
$t_{rise} = \ln(9) \times 0.5ns = 1.098 ns$	1.103ns
$t_{fall} = \ln(9) \times 0.5ns = 1.098 ns$	1.101ns

4) Do parametric sweep for $R = 1:1:5k\Omega$. Report overlaid results. Comment on the results

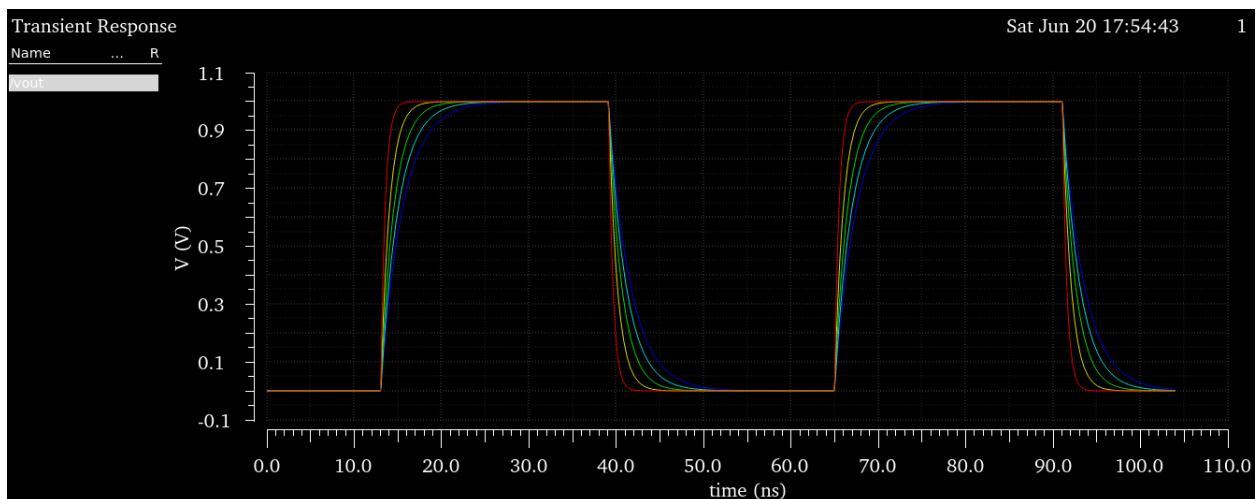


Figure 4: output parametric sweep



Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: R=1k						
1	lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	1.103n			
1	lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	1.101n			
Parameters: R=2k						
2	lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	2.198n			
2	lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	2.201n			
Parameters: R=3k						
3	lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	3.277n			
3	lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	3.289n			
Parameters: R=4k						
4	lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	4.388n			
4	lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	4.375n			
Parameters: R=5k						
5	lpf_part1	riseTime(VT("/vout") 0 nil 1 nil 1...	5.485n			
5	lpf_part1	fallTime(VT("/vout") 0 nil 1 nil 90...	5.482n			

Figure 5: t_{rise} & t_{fall} when change R 1--> 5k

2. AC Analysis

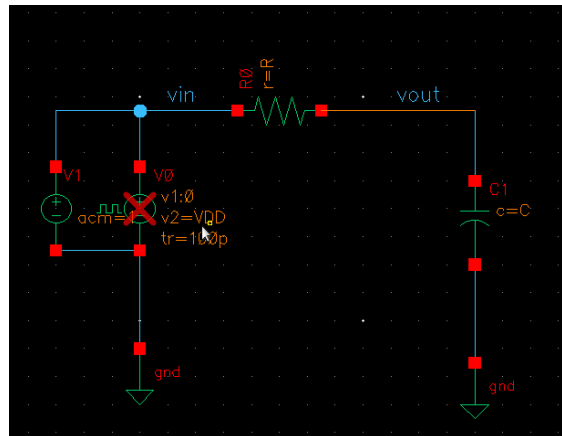


Figure 7: AC Analysis



1) Report Bode Plot (magnitude and phase) for the previous LPF

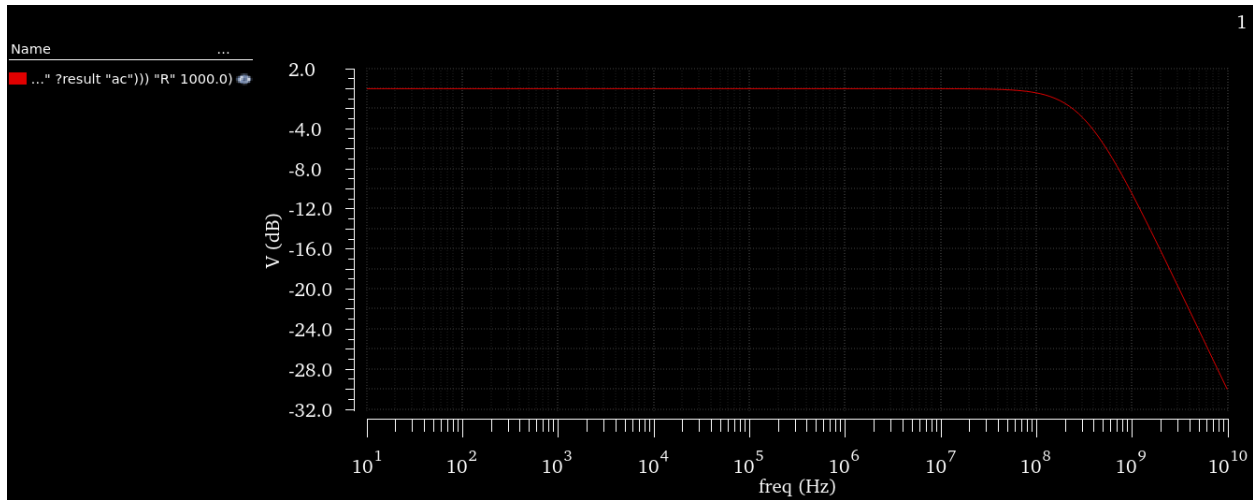


Figure 8: $20\log(v_{out})$ vs freq

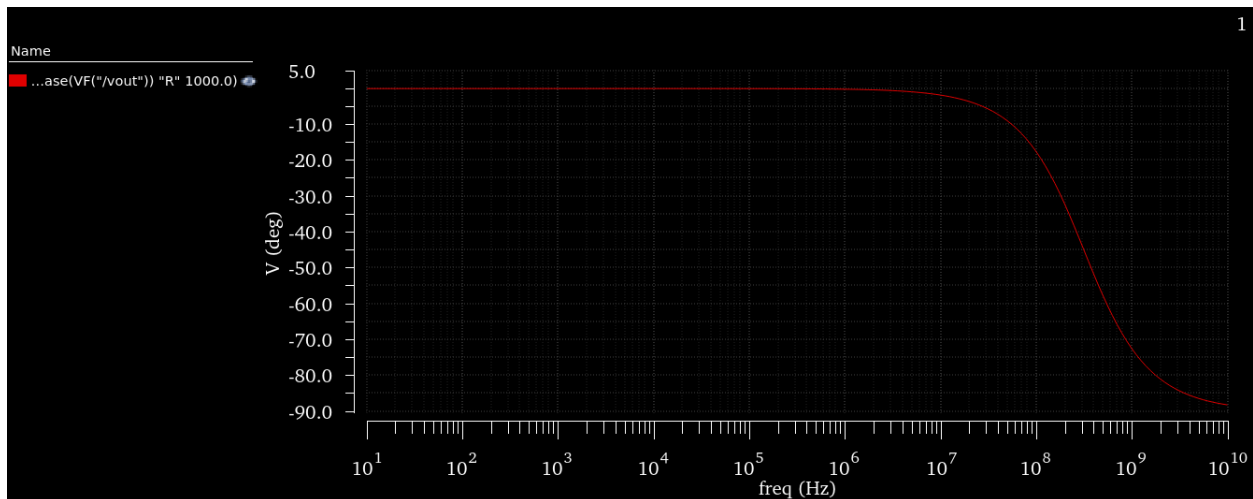


Figure 10: phase



2) Calculate DC gain and 3dB bandwidth using Cadence calculator expressions.
Export the expressions to adexl.

Test	Output	Nominal	Spec	Weight	Pass/Fail
lpf_part1	/vout				
lpf_part1	/vin				
lpf_part1	BW	317.6M			
lpf_part1	Ao	1			

Figure 11: BW & DC gain

3) Compare simulation with analytical results in a table

Parameter	simulation	analytical
DC Gain	1=0db	1 = 0 db
3dB Bandwidth	317.6 MHz	$f_{3dB} = \frac{1}{2\pi \times 0.5 \times 10^{-9}} = 318.309 \text{ MHz}$



4) Do parametric sweep for $R = 1, 10, 100, 1000 k\Omega$. Report overlaid results.
Comment on the results

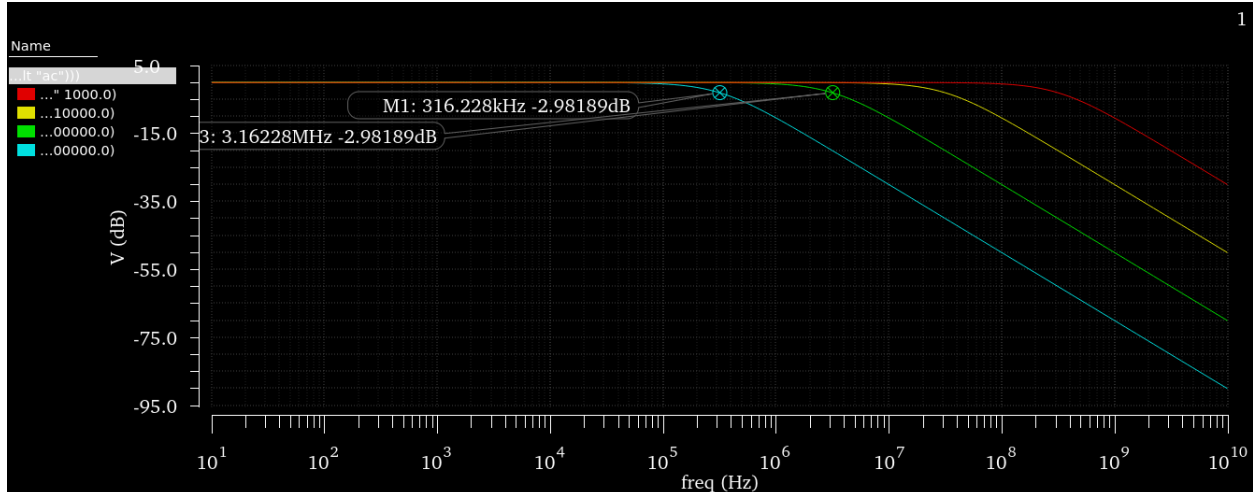


Figure 12: parametric sweep for $R = 1, 10, 100, 1000 k\Omega$

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: R= 1k						
1	lpf_part1	/vout				
1	lpf_part1	/vin				
1	lpf_part1	BW	317.6M			
1	lpf_part1	Ao	1			
Parameters: R= 10k						
2	lpf_part1	/vout				
2	lpf_part1	/vin				
2	lpf_part1	BW	31.76M			
2	lpf_part1	Ao	1			
Parameters: R= 100k						
3	lpf_part1	/vout				
3	lpf_part1	/vin				
3	lpf_part1	BW	3.176M			
3	lpf_part1	Ao	1			
Parameters: R= 1M						
4	lpf_part1	/vout				
4	lpf_part1	/vin				
4	lpf_part1	BW	317.6k			
4	lpf_part1	Ao	1			

Figure 14: change $R \rightarrow$ change BW



Phase vs R :

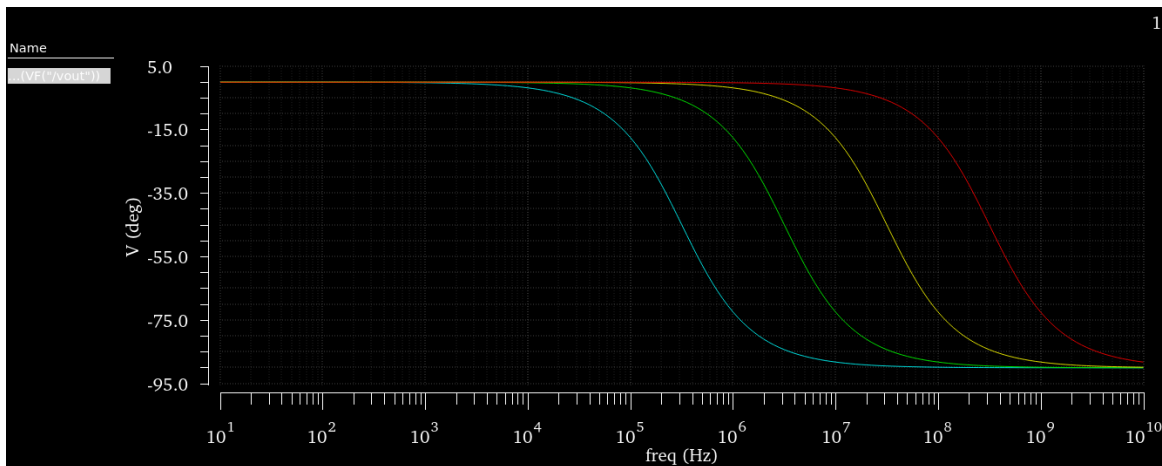


Figure 15: phase vs R

Comment on the Overlaid AC Analysis Results

- **Effect on the 3dB Bandwidth:** As the resistance R increases, the 3dB cut-off frequency (f_{3dB}) decreases. This relationship is governed by the analytical formula:

$$f_{3dB} = \frac{1}{2\pi RC}$$

- Because the values of R are swept in decades (1 kΩ, 10 kΩ, 100 kΩ, 1000 kΩ), the bandwidth is reduced by a factor of 10 for each step. This visually shifts the magnitude curve to the left by one full decade per step on the frequency axis.
- **Effect on the DC Gain:** The DC gain remains constant at 0 dB (1 V/V) across all values of R. At DC (zero frequency), the capacitor acts as an open circuit, meaning no steady-state current flows through the resistor. Therefore, there is no voltage drop across R, making the output voltage equal to the input voltage regardless of the resistor's magnitude.
- **Effect on the Phase Response:** Consistent with the magnitude curve, the phase response also shifts to the left as R increases. For every resistor value in the parametric sweep, the phase angle crosses exactly -45° at its new corresponding f_{3dB} cut-off frequency.



3. Pole Zero Analysis

1) Report pole zero analysis results.

qfactor	/poles R 1000 (Hz)
1 500.0E-3	318.3E6

2) Find the pole frequency and compare it with the bandwidth calculated from AC analysis.

The pole-zero analysis was performed using Cadence. The simulation result shows that the first-order low pass filter has a single pole located at 318.3 MHz



Part 2: MOSFET Characteristics

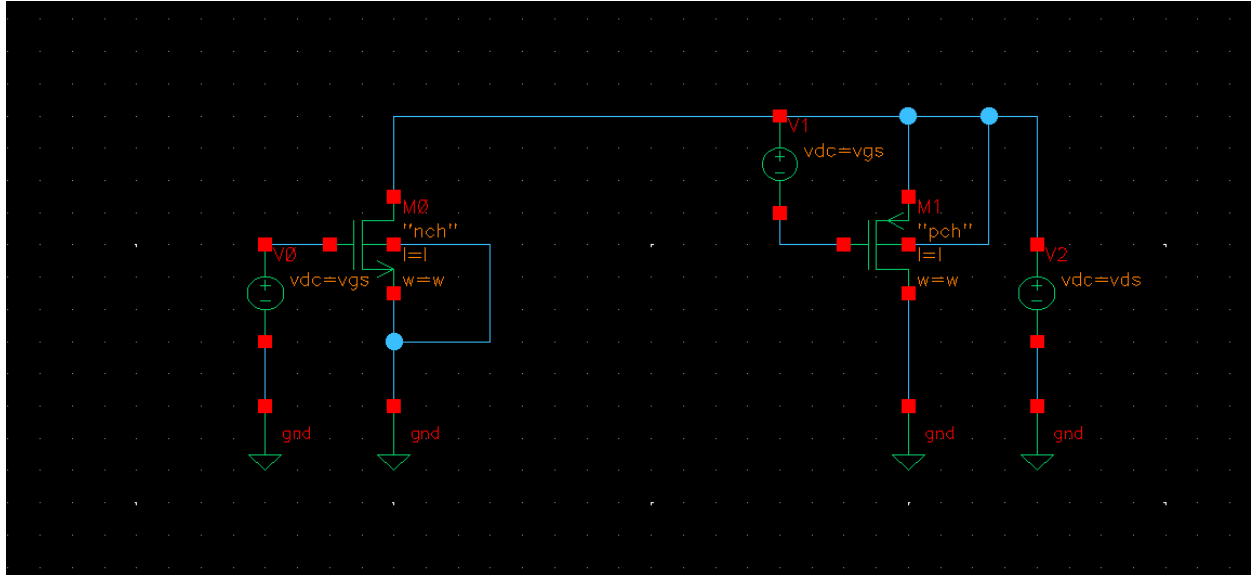


Figure 16: nmos & pmos schematic

- Using random aspect ratio $= \frac{W}{L} = 75$
- L is parameter $\rightarrow W = 75 * L$

1. ID vs VGS

1) Plot $I_D - V_{GS}$ characteristics for NMOS and PMOS devices.

- nmos

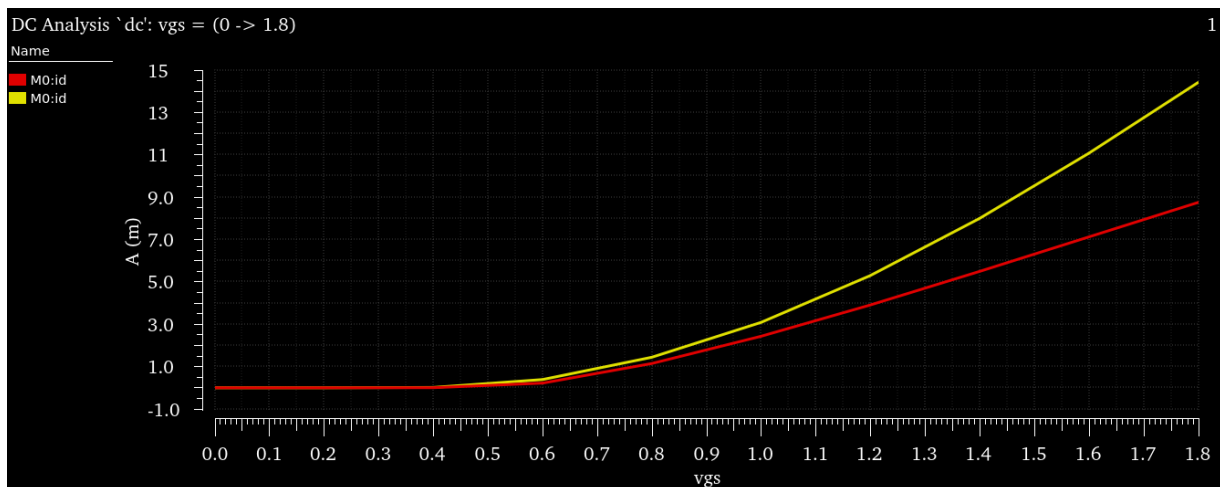


Figure 17: nmos_short channel(red)_ long_channel(yellow)

- pmos

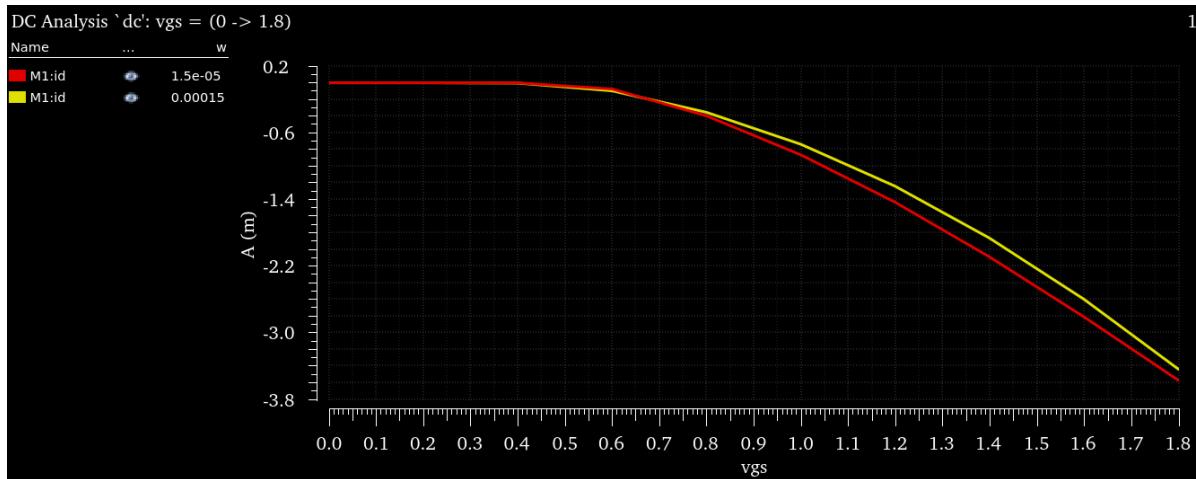


Figure 18: pmos_short channel(red)_long channel(yellow)

2) Comment on the differences between short channel and long channel results.

- Which one has higher current? Why?

→ Long channel; because it is not limited by velocity saturation. In short-channel devices, the strong electric field causes carriers to reach their maximum velocity quickly, limiting the current to a linear relationship with VGS. The long-channel device, however, maintains the traditional square-law behavior, allowing its current to grow quadratically and reach higher values for the same W/L ratio.

- Is the relation linear or quadratic? Why?

→ linear in short channel & quadratic in long channel; because short-channel devices suffer from velocity saturation at high electric fields, which limits the current growth to a linear rate. In contrast, long-channel devices operate under lower electric fields, avoiding velocity saturation and maintaining the ideal quadratic square-law behavior.

$$E = \frac{V_{ds}}{L}, v_{drift} \propto E \text{ (until reach to } v_{drift} \text{ saturation)}$$



3) Comment on the differences between NMOS and PMOS.

- Which one has higher current? Why?

→ The **NMOS** device exhibits a higher drain current. This is because the mobility of electrons (μ_n) is significantly higher (typically 2 to 3 times) than the mobility of holes (μ_p). Since the drain current is directly proportional to carrier mobility, the NMOS delivers more current for the same aspect ratio (W/L) and bias conditions

- What is the ratio between NMOS and PMOS currents at $V_{GS} = V_{DD}$?

→ short channel nmos & pmos :

$$I_n/I_p = 8.7442/3.5685 = 2.45$$

Because mobility of electrons

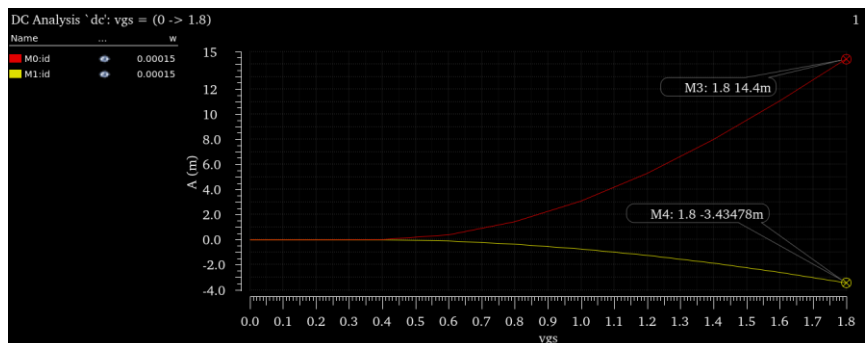
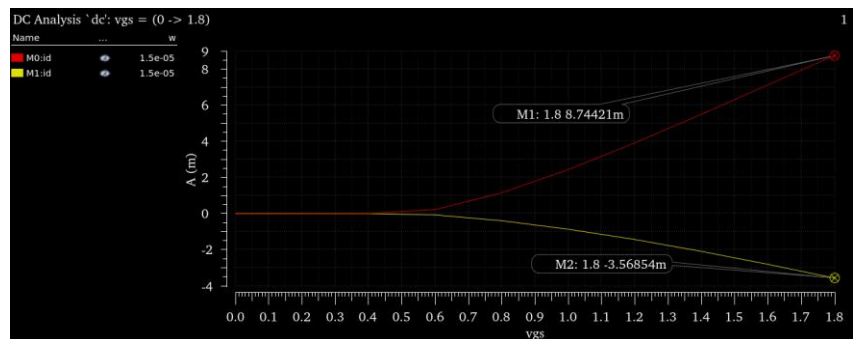
(Charge carriers in nmos) >

mobility of holes

(Charge carriers in pmos).

→ long channel nmos & pmos :

$$I_n/I_p = 14.4/3.4347 = 4.2$$



Note:

The current ratio (I_n/I_p) decreases in the short-channel device compared to the long-channel device due to the **velocity saturation** effect. In the long-channel device, both transistors follow the ideal square-law model, and the ratio (4.2) directly reflects the intrinsic mobility ratio (μ_n/μ_p). However, in the short-channel device, because electrons possess higher mobility than holes, they reach their

saturation velocity (v_{sat}) at much lower electric fields. Consequently, the NMOS device is more severely affected by velocity saturation earlier than the PMOS. This limits the NMOS current growth to a linear behavior, significantly degrading its current and causing the overall ratio to drop to (2.45).

- Which one is more affected by short channel effects?

→ The **NMOS** is more severely affected by short-channel effects, particularly velocity saturation. Because electrons have a significantly higher mobility than holes, they reach their maximum saturation velocity (v_{sat}) at much lower electric fields. Consequently, the NMOS device deviates from the ideal square-law behavior and experiences current degradation earlier and more severely than the PMOS device for the same channel length.

2. g_m vs V_{GS}

1) Plot g_m vs V_{GS} for NMOS device.

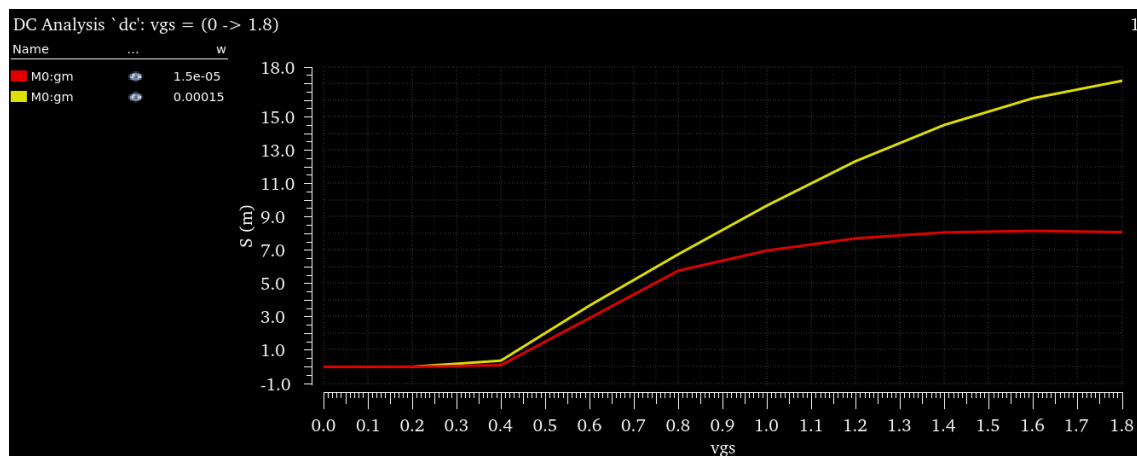


Figure 19: short(red) vs long(yellow)

2) Comment on the differences between short channel and long channel results.

- Does g_m increase linearly? Why?

→ Yes, in the **long-channel device**, g_m increases linearly with V_{GS} . This occurs because the device follows the ideal square-law model, where I_D is proportional to $(V_{GS} - V_{TH})^2$. Since g_m is the derivative of I_D with respect to V_{GS} ($g_m =$



$\partial I_D / \partial V_{GS}$), differentiating this quadratic relationship yields a linear dependence of g_m on V_{GS} ($g_m \propto V_{GS} - V_{TH}$)

- Does g_m saturate? Why?

→ Yes, g_m saturates, but specifically in the **short-channel device** at higher V_{GS} values. This is fundamentally due to **velocity saturation**. As the electric field increases, charge carriers reach their maximum saturation velocity (v_{sat}), causing the drain current I_D to transition from a quadratic to a linear dependence on V_{GS} . Because the derivative of a linear function is a constant, the transconductance g_m flattens out and becomes saturated (constant) at high V_{GS} values, approaching $g_m \approx WC_{ox}v_{sat}$.

3. I_D vs V_{DS}

1) Plot $I_D - V_{DS}$ characteristics for NMOS device.

→ long channel

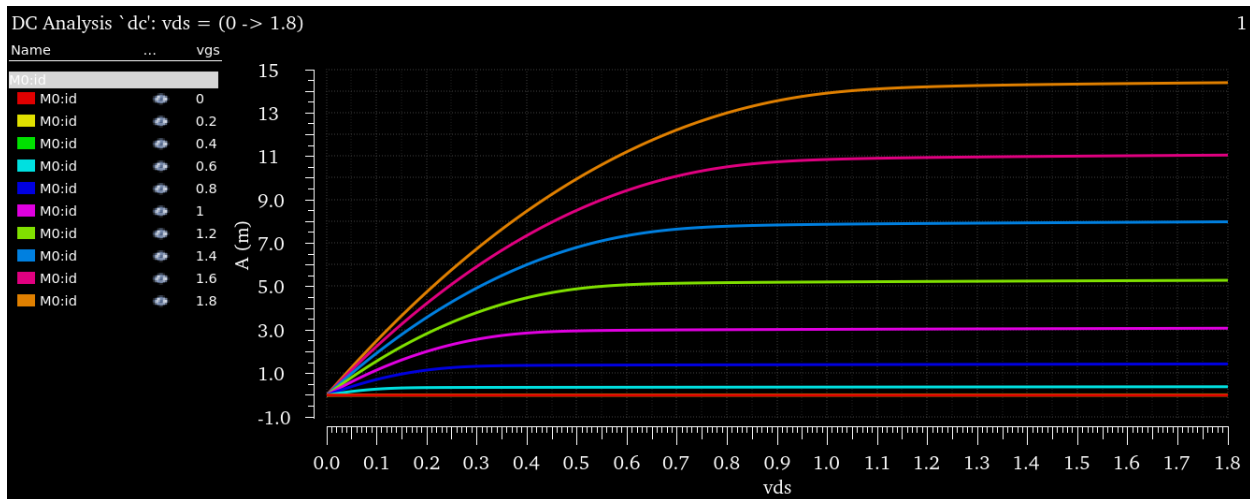


Figure 20: I_D vs v_{ds} long channel

→ short channel

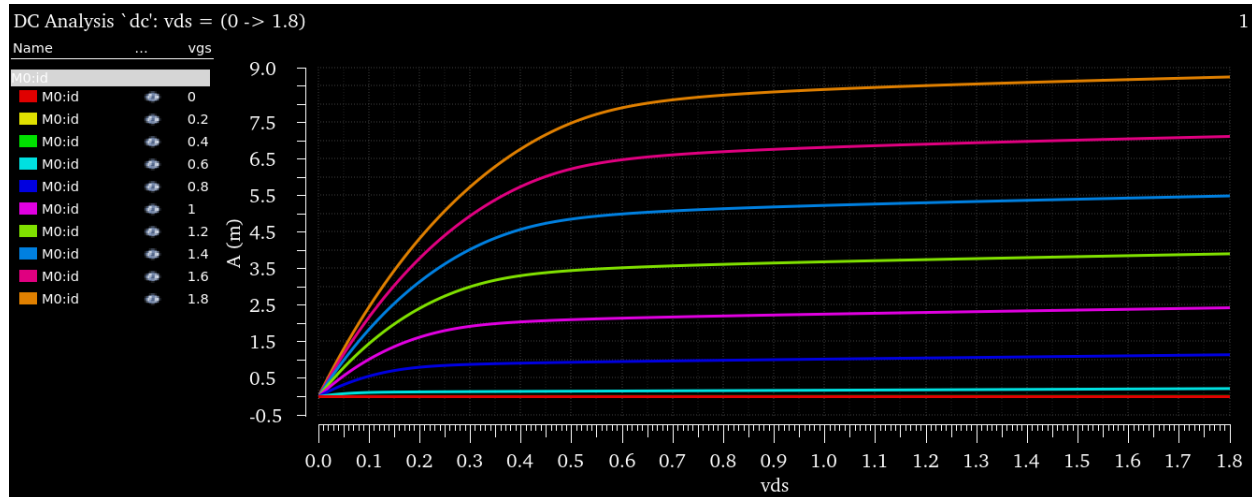


Figure 21: I_d vs v_{ds} long channel

2) Comment on the differences between short channel and long channel results.

- Which one has higher current? Why?

→ The **long-channel device** exhibits a higher drain current. As previously analyzed, the short-channel device suffers from velocity saturation, which limits its current growth to a linear dependence on the overdrive voltage. In contrast, the long-channel device does not experience significant velocity saturation, allowing its current to increase quadratically and reach higher values for the same bias conditions and W/L ratio.

- Which one has higher slope in the saturation region? Why?

→ The **short-channel device** exhibits a higher slope in the saturation region. This is caused by the **Channel Length Modulation (CLM)** effect. As V_{DS} increases in saturation, the pinch-off point moves towards the source, reducing the effective channel length by a distance ΔL . In a short-channel device, this ΔL represents a significant fraction of the total length L , leading to a noticeable increase in I_D (a higher slope). In the long-channel device, ΔL is negligible compared to the large overall length L , resulting in an almost perfectly flat curve ($slope \approx 0$).

Consequently, the short-channel device has a much larger channel length modulation parameter ($\lambda \propto 1/L$) and a lower output resistance (r_o).



4. [Optional] g_m and r_o in Triode and Saturation

1) Plot g_m and r_o vs V_{DS} for NMOS device

→ g_m vs V_{ds}

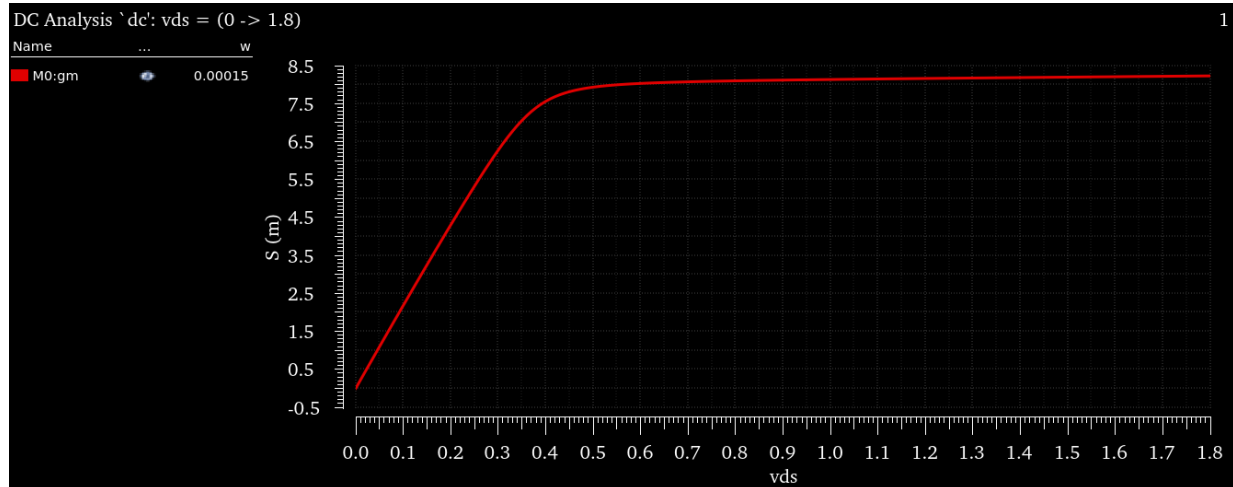


Figure 22: g_m vs v_{ds}

→ r_o vs V_{ds}

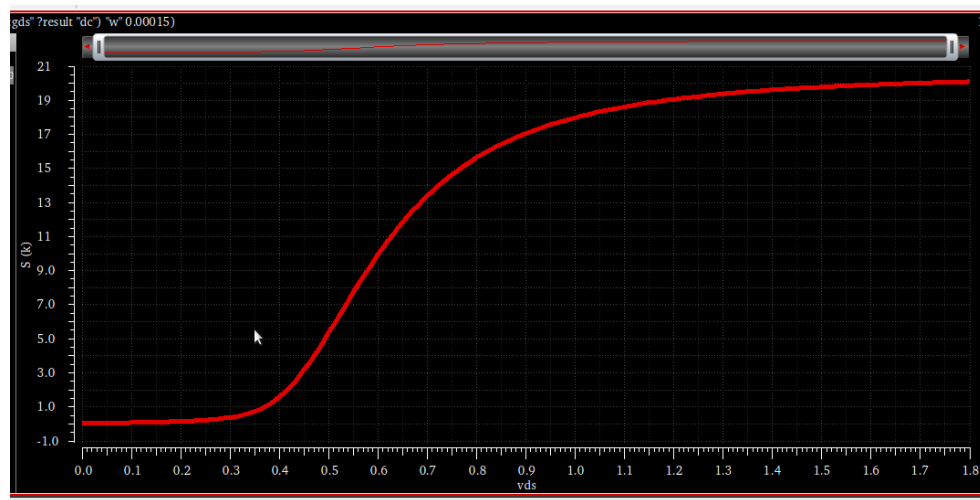


Figure 23: r_o vs v_{ds}



2) Comment on the variation of g_m vs V_{DS} .

- In the first part of the curve, is the relation linear? Why?

→ **Yes, the relation is linear in the first part of the curve.** In this region, the transistor operates in the **triode region**. According to the device physics for the triode region, the transconductance is directly proportional to the drain-to-source voltage ($g_m = \mu_n C_{ox} \frac{W}{L} V_{DS}$). Therefore, increasing V_{DS} leads to a linear increase in g_m .

- Does g_m saturate? Why?

→ **Yes, g_m saturates.** As V_{DS} increases and exceeds the overdrive voltage ($V_{DS} \geq V_{GS} - V_{TH}$), the transistor enters the **saturation region**. In saturation, the drain current becomes relatively independent of V_{DS} , and the transconductance becomes a function of V_{GS} only ($g_m \approx \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$). Since V_{GS} is held constant during this sweep, g_m flattens out and reaches a constant saturated value.

- Where do you want to operate the transistor for analog amplifier applications? Why?

→ For analog amplifier applications, the transistor must be operated in the saturation region. As observed in the g_m vs V_{DS} curve, the transconductance (g_m) reaches its maximum and saturated value in this region. Since the intrinsic voltage gain of an amplifier is directly proportional to transconductance ($A_v = g_m \times r_o$), operating in saturation provides the highest possible gain. Furthermore, the constancy of g_m in this region ensures linear amplification, minimizing distortion as the output voltage (V_{DS}) swings during operation.



3) Comment on the variation of r_o vs V_{DS} .

- **Does r_o saturate just after the transistor enters saturation similar to g_m ? Why?**
 - ➔ **No, r_o does not saturate immediately.** Unlike g_m , which stabilizes quickly once $V_{DS} \geq V_{DS,sat}$, r_o continues to rise gradually. This behavior is governed by the **Channel Length Modulation (CLM)** effect. As V_{DS} increases beyond the pinch-off point, the effective channel length decreases progressively, causing the output resistance to increase smoothly and gradually rather than abruptly.
- **Does r_o increase if the transistor is biased more into saturation?**
 - ➔ **Yes, it does.** As the transistor is biased deeper into the saturation region (higher V_{DS}), the output resistance r_o increases significantly until it eventually reaches its maximum saturated value.
- **Should we operate the transistor at the edge of saturation?**
 - ➔ **No, we should avoid operating at the edge of saturation.** At the boundary between the triode and saturation regions ($V_{DS} \approx V_{DS,sat}$), r_o is still in its transition phase and has a relatively low value. Operating here would severely degrade the performance of the amplifier.
- **Where do you want to operate the transistor for analog amplifier applications? Why?**
 - ➔ For analog amplifier applications, the transistor must be operated **deep in the saturation region**. The intrinsic voltage gain of an amplifier is directly proportional to both transconductance and output resistance ($A_v = g_m \times r_o$). Operating deep in saturation ensures that r_o reaches its maximum possible value, which, combined with a saturated g_m , provides the highest possible voltage gain.